FreeFlow-ONT User Manual

(version 1.0)

- FreeFlow-ONT is a Galaxy-based framework for the analysis of Oxford Nanopore Technologies direct RNA sequencing (ONT DRS) data. It provides an integrated and user-friendly environment that supports key steps of ONT DRS analysis, including data preprocessing, alignment, transcript-level analysis, and downstream functional exploration. The framework is designed to improve accessibility, standardization, and efficiency in ONT DRS data analysis for both routine and advanced research applications.
- FreeFlow-ONT was powered with an advanced packaging technology, which enables compatibility and portability.
- FreeFlow-ONT project is hosted on https://github.com/jy-ai/FreeFlow-ONT
- FreeFlow-ONT docker image is available at https://hub.docker.com/r/malab/freeflowont

Data Preparation Module

This module prepares raw Oxford Nanopore Technologies (ONT) sequencing data for downstream analysis. It includes two tools: **FAST5 to SLOW5 Conversion** and **Basecalling**. The first tool converts raw FAST5 files into SLOW5/BLOW5 format to improve data accessibility and computational efficiency, while the second tool translates raw electrical signals into nucleotide sequences in FASTQ format.

Tools	Description	Input	Output	Time (test data)	Reference
FAST5 to SLOW5	Convert raw ONT FAST5 files into SLOW5/BLOW5 format for efficient downstream analysis	FAST5 files	SLOW5/BLOW5 files	Depends on the file size	slow5tools
Basecalling	Convert raw nanopore signal data into nucleotide sequences	FAST5 or SLOW5/BLOW5 files	FASTQ files	Depends on the dataset size and hardware configuration	ONT basecaller

FAST5 to SLOW5

This function is designed to convert raw ONT sequencing data from **FAST5** format into **SLOW5** or **BLOW5** format. Compared with FAST5, SLOW5/BLOW5 provides a more efficient and lightweight format for raw nanopore signal storage and can improve the performance of downstream analysis tools.

Input

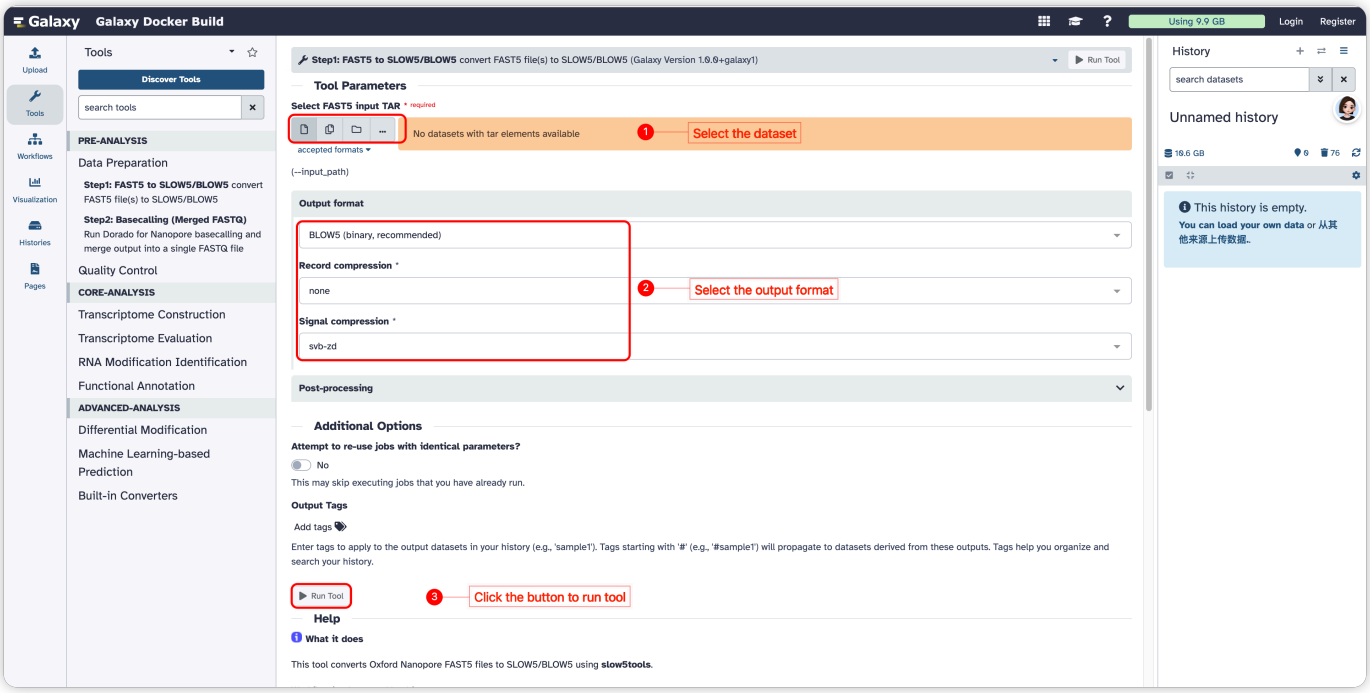
- **Input FAST5 files:** Raw ONT sequencing files in FAST5 format.
- **Input type:** Single-FAST5 files or a directory/archive containing multiple FAST5 files.

Parameters

- **Output format:** Select the output format as **SLOW5** or **BLOW5**.
- **Compression mode:** Optional, specify whether compressed output should be generated.

Output

- **Converted SLOW5/BLOW5 file(s)** for downstream analysis.



Basecalling

This function is designed to convert raw nanopore electrical signals into nucleotide sequences through basecalling. The generated sequence reads are saved in **FASTQ** format and can be used for downstream analyses such as alignment, transcript quantification, and modification detection.

Input

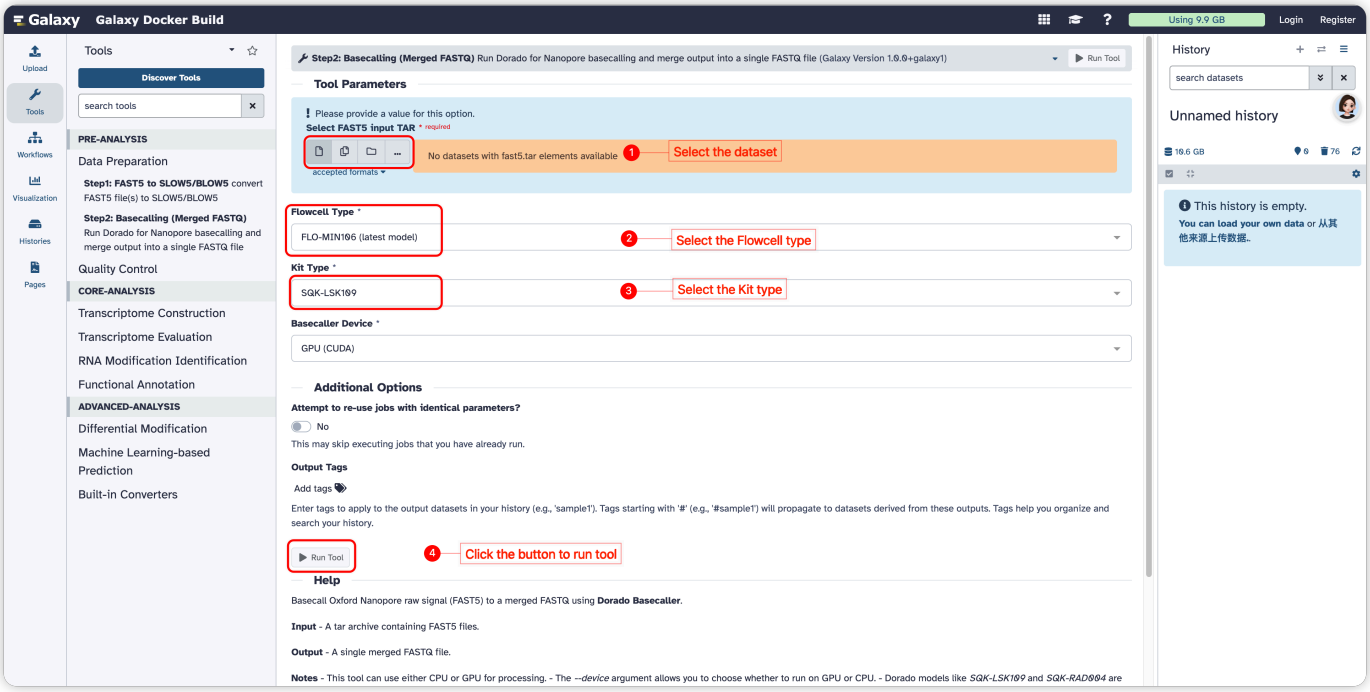
- **Input raw signal files:** FAST5 or SLOW5/BLOW5 files generated from ONT sequencing.
- **Model file or model selection:** A basecalling model specified by the selected from the available presets.

Parameters

- **Basecalling model:** Select the appropriate model for RNA or DNA data, depending on the sequencing type.
- **Device:** Choose CPU or GPU for basecalling.

Output

- **Basecalled reads in FASTQ format**



Quality Control Module

This module is designed to improve the quality of ONT sequencing reads before downstream analysis. It currently includes the **Chopper Quality Control** tool, which filters reads according to user-defined quality and length thresholds. An optional **NanoPlot** report can also be generated for visual quality assessment.

Tools	Description	Input	Output	Time (test data)	Reference
Chopper Quality Control	Filter ONT reads by minimum quality and read length, with optional NanoPlot QC visualization	FASTQ file	Filtered FASTQ file; optional NanoPlot HTML report	Depends on the dataset size	Chopper; NanoPlot

Chopper Quality Control

This function is designed to filter Oxford Nanopore sequencing reads in **FASTQ** format using **Chopper**. Reads can be retained according to user-defined thresholds for minimum base quality and minimum read length. In addition, users may optionally run **NanoPlot** to generate a visual quality control report for the filtered reads.

Input

- **Input FASTQ file:** A FASTQ file containing ONT sequencing reads.

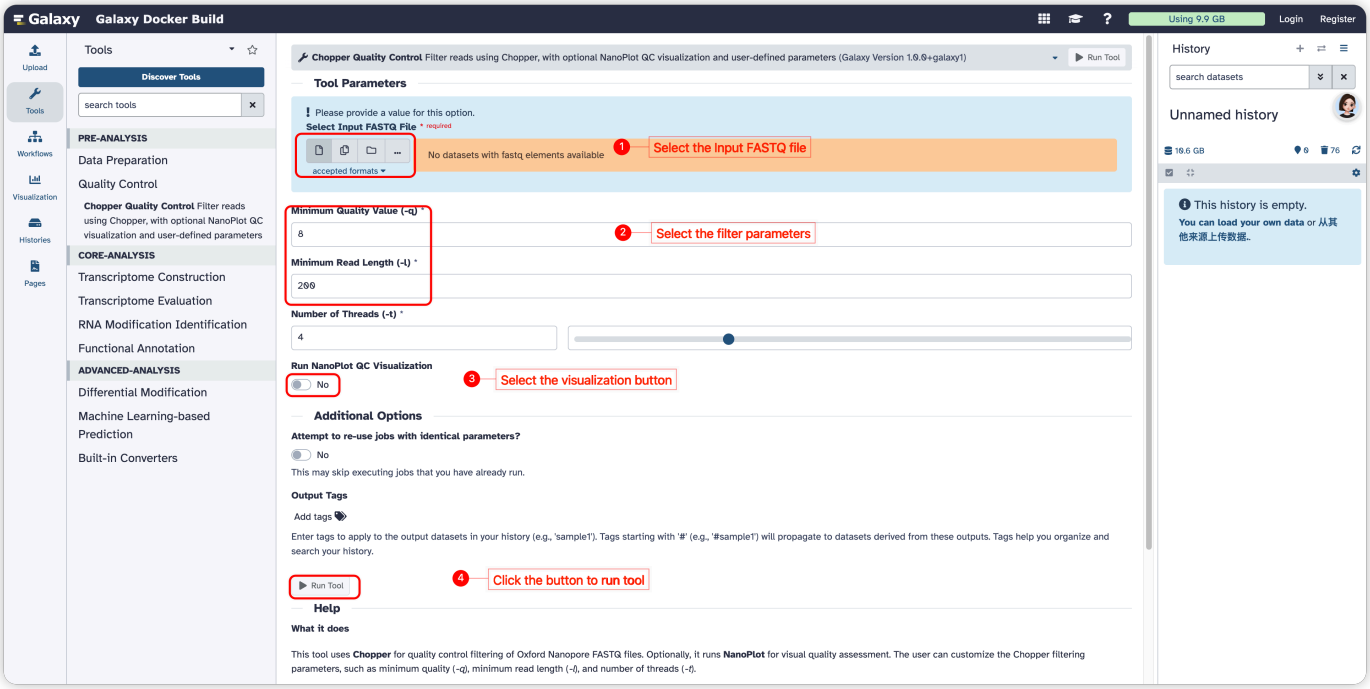
Parameters

- **Minimum Quality Value (-q):** An integer specifying the minimum average read quality required for a read to be retained. The default value is **8**.
- **Minimum Read Length (-l):** An integer specifying the minimum read length required for a read to be retained. The default value is **200**.

- **Number of Threads (-t):** An integer specifying the number of CPU threads used during filtering. The default value is **4**.
- **Run NanoPlot QC Visualization:** Select whether to generate an additional **NanoPlot** HTML report for visual inspection of the filtered reads.

Output

- **Filtered FASTQ file:** A FASTQ file containing reads that pass the user-defined quality and length thresholds.
- **NanoPlot report (optional):** An HTML report summarizing read length and quality distributions of the filtered dataset when NanoPlot visualization is enabled.

Notes

- **Chopper** is used to remove low-quality and short reads from ONT FASTQ files.
- The filtering process is controlled by user-defined thresholds for quality and read length.
- When enabled, **NanoPlot** provides graphical summaries that help users evaluate the quality of the filtered reads.